

AUTHENTICATION LOGS — Windows Security · SSH · Zeek · Entra ID · Okta · CSV / JSONL · LANL

log lines · JSON · batches

1 INGESTION

Source adapters

7 log formats → one event shape:
time · account · from host · to host · success

Entity dictionary

a stable number for every account and host,
frozen together with the model checkpoint

normalised events

2 FEATURES
& MODEL

Causal feature engine

27 features per event,
computed from the past only

Temporal Graph Network

64-number memory per node;
score first, then update

Warm state

memory + counters saved and
restored: a restart loses nothing

27 features

P(malicious) for every event

3 DETECTION

Signal tracker

novelty · burst · pivot
explicit, explainable

Chain rule

4 hops into new hosts
within 30 minutes

Fusion + gate

noisy-OR of all sources
alert 0.33 · auto 0.85

ATT&CK attribution

technique + confidence
+ evidence bundle

alerts +
evidence

alert: technique · confidence · risk

4 RESPONSE

Response coordinator

one execution path; budget
20 per hour · 1 per account

Prevention loop

contain → verify 30 min →
escalate (lock) → revert 60 min

Executor + connectors

the exact command and its undo;
dry-run unless armed

action
records

approve ·
keep · lift

real commands only when armed · irreversible actions never run unattended

DIRECTORY · ENDPOINT · NETWORK — Active Directory / Entra ID · EDR isolate · host firewall

5 CONSOLE · API · STORE

Analyst console

11 workspaces: alerts,
paths, cases, response

REST API

72 routes · API key
health · metrics

SQLite store

alerts · actions · audit
replayed on restart



SOC analyst

investigates · approves
keeps or lifts a lock

One Python process · one ordered event stream · live scores equal offline scores · every action is recorded with the command that ran